MULTI REGRESSION WITH A BINARY DEPENDENT VARIABLE

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Regression with a binary dependent variable

1. Linear Probability Model (LPM)
2. Logit Model
3. Probit Model
4. Comparing LPM, Logit, Probit

I. Linear Probability Model (LPM)

- LPM is used when the dependent variable is a dummy/ binary variable.
- As Y has only 2 values (0 or 1), β_j can not interpreted as the change in Y when X changes 1 unit, ceteris paribus;
- Y changes from value 0 to 1.
- However, β_j can still be interpreted.

1.Model: Probability analysis

We have a model:

$Yi = \beta 0 + \beta 1X1i+ui$

Let pi be probability for event A happening given Xi, pi = P(A/Xi),

So, probability for event A not happening given Xi: $P(\overline{A}) = 1\text{-}pi$

Model: $Y_i = \beta_0 + \beta_1 X_i + u_i$ *(1.1)*

Where: Y- dependent variable (qualitative)

$$Y_i = \begin{cases} 1 & \text{, if event A happens} \\ 0 & \text{, if event A doesn't happen} \end{cases}$$

We can write:

$p_i = P(Y=1|X_i)$; $1-p_i = P(Y=0|X_i)$

- Linear regression model: when X changes by 1 unit, the average value of Y (E (Y|X)) changes by β1 unit.
- LPM: when X changes by 1 unit, the probability for the event A happening changes by β1 unit.

➔ **E(Yi/Xi) = P(Y=1/Xi) = β0 + β1X1= pi = 1**

We can write the regression model as:

+ $P(y=1|x) = \beta 0 + \beta 1Xi$ as a linear function of X

➔ Regression model with a binary dependent variable is called Linear Probability Model (LPM) as $P(y=1|x)$ is linear with parameters βj

How to interpret coefficients in LPM

+ Apply the OLS to estimate coefficients
+ The same assumptions of the linear regression model
+ βj shows the change in probability of y = 1 (probability of event A happening) when Xj changes 1 unit.
+ $\beta 0$ shows the change in probability of y = 1 when Xj = 0.

For example

inlf = 0,5 + 0,038educ – 0,02female +u^

+ Holding other factors fixed, another year of education will increase the probability of attending labor force by 3,8%

+ Holding other factors fixed, female has lower probability of attending labor force by 2%.

Weakness of LPM

1. Probability can be smaller than 0 or greater than 1.
2. Probability is linear with independent variable. But in fact, it cannot be.

Exercise 1

A survey of 40 students after graduation for 6 months, with variables GS - Graduate Score, EN – English grade. The scale of grade equals 100.

Y = 1, if student can have a job, Y =0, if student has not got a job yet. (inlf)

Let "probability of getting job" is the probability for a student to get a job after 6 month from graduation, α =5%

Variable	Coefficient	Std. Error	t-Statistic	Prob
GS	0.020158	0.00835	2.414214	0.0208
EN	0.02922	0.009462	3.088114	0.0038
_const	-3.01567	0.54346	-5.549015	0.0000

1. Write LPM.
2. Interpret coefficients
3. Estimate the probability of getting job when GS and EN equal: (70,80); (60,60)?

Answer

1. LPM

$$p^{LPM} = P(Y=1 \mid X) = E(Y / GS, EN) = \beta_0 + \beta_1 GS_i + \beta_2 EN_i$$

$$\hat{p}^{LPM} = -3.01567 + 0.020158 GS_i + 0.02922 EN_i$$

$$Y = -3.01567 + 0.020158 GS_i + 0.02922 EN_i + Ui \wedge$$

2. Interpretation:

- P-value <0.05 ➔ both GS and EN have statistically significance on Y.
- Holding other factors fixed, when GS increases by 1 unit, the probability of getting a job increases by 2%.
- Holding other factors fixed, when EN increases by 1 unit, the probability of getting a job increases by 2.9%.

3. When GS = 70; EN = 80:

$$\hat{p}^{LPM}(Y=1|X) = -3.01567 + 0.020158*70 + 0.02922*80 = 0.733$$

+ When GS =60; EN =60:

$$\hat{p}^{LPM}(Y=1|X) = -3.01567 + 0.020158*60 + 0.02922*60 = -0.053$$

The negative value is inappropriate. So we can not have a conlusion in this case.

II. Logit and Probit Model

As LPM has 2 weaknesses:

+ P(y=1|x) can be smaller than 0 or greater than 1

+ the effect of X on Y is constant

To overcome these weaknesses, we rewrite the model:

$$P(y=1|x) = G(\beta_0 + \beta_1 X_1 + \ldots + \beta_k X_k) = G(z)$$

+ If we want G(z) to have value in the interval (0,1) => Logit and Probit model are among the choices.

Logit Model

$$G(z) = \exp(z) / [1 + \exp(z)]$$

+ In the logit model, G(z) is the logistic function which is between 0 and 1 for all real numbers z.
+ This is the cdf for a standard logistic random variable.
+ P(y=1|x) = G(z) = pi and P(y=0|x) = 1- G(z) = 1- pi

The effect of Xi on P(y=1|x) ?

$$\frac{\partial p(x)}{\partial x_j} = \frac{\beta_j \exp^z}{\left(1+\exp^z\right)^2} = p_i(1-p_i)\beta_j$$

At each value of Xi, the probability for the event A happening is pi. When X changes by 1 unit, the probability changes: pi(1-pi)βj.

2. Ước lượng mô hình Logit

a.Phương pháp Berkson

Berkon biến đổi mô hình Logit qua hàm Logarit tỷ lệ ưu thế:

$$L_i = \ln\left(\frac{p_i}{1-p_i}\right) + u_i = \ln\left(e^{\beta_1 + \beta_2 X_i}\right) + u_i = \beta_1 + \beta_2 X_i + u_i$$

Giả sử rằng mẫu có Ni giá trị Xi; trong Ni quan sát này chỉ có ni giá trị mà Yi =1, khi đó ước lượng điểm của pi là:

$$\hat{p}_i = \frac{n_i}{N_i}$$

Chúng ta dùng ước lượng pi để ước lượng mô hình:

$$\hat{L}_i = \ln\left(\frac{\hat{p}_i}{1-\hat{p}_i}\right) = \hat{\beta}_1 + \hat{\beta}_2 X_i$$

Thủ tục ước lượng:

Bước 1: Với mỗi Xi ta tính $\hat{p}_i = \frac{n_i}{N_i}$; $\hat{L}_i = \ln\left(\frac{\hat{p}_i}{1-\hat{p}_i}\right)$

$$\hat{w}_i = N_i \hat{p}_i (1-\hat{p}_i)$$

Bước 2: Thực hiện biến đổi biến số

$$\sqrt{w_i} L_i = \beta_1 \sqrt{w_i} + \beta_2 \sqrt{w_i} X_i + \sqrt{w_i} u_i$$

OLS mô hình trên thu được $\hat{\beta}_1$ $\hat{\beta}_2$

Nhược điểm: Mẫu phải rất lớn.

Khi mẫu không đủ lớn; phải dùng phương pháp ước lượng hợp lý tối đa.

b. Phương pháp ước lượng hợp lý tối đa

- Do hàm E(y|x) là không tuyến tính nên phương pháp OLS không còn hợp lý.
- Phương pháp ước lượng hợp lý tối đa (maximum likelihood estimation) hợp lý hơn do dựa trên sự phân phối có điều kiện của y.
- Hàm mật độ có điều kiện của y:

$f(y=1|x;\beta) = P(y=1|x) = G(x\beta)$

Và $f(y=0|x;\beta) = P(y=0|x) = 1-G(x\beta)$

Chúng ta có thể viết lại như sau:

$$f(y \mid x_i;\beta) = [G(x_i\beta)]^y [1 - G(x_i\beta)]^{1-y}, y = 0,1$$

Hàm logarit của f(y|x;β) là:

$$\ell(\beta) = y\log[G(x_i\beta)] + (1\text{-}y)\log[1 - G(x_i\beta)]$$

Tìm MLE của β, được ký hiệu là β, nhằm tối đa tổng của hàm logarit trên.

Tìm β để $\sum \ell(\beta)\max$

Probit Model

1. Model

$$P(y=1|x)=G(\beta_0+\beta_1 X_1+\ldots+\beta_k X_k)=G(z)$$

G is standard normal cdf, which is expressed as an integral:

$$G(z)=\Phi(z)=\int_{-\infty}^{z}\frac{1}{\sqrt{2\pi}}\exp(-z^2/2)dz$$

When X changes by 1 unit, P(y=1|x) changes:

$$\frac{\partial p(x)}{\partial x_j}=\beta_j[(2\pi)^{-1/2}\exp(-z^2/2)]$$

+ Some properties of cdf:

- The values is between (0;1)
- $F(0) = 0.5$; $F(-\infty) = 0$; $F(+\infty) = 1$

+ Estimation method: MLE

Exercise

A survey of 40 students after graduation for 6 months, with variables GS - Graduate Score, EN – English grade. The scale of grade equals 100.

Y = 1, if student can have a job, Y =0, if student has not got a job yet.

Let “probability of getting job” is the probability for a student to get a job after 6 month from graduation, α =5%

Exercise 1: Estimation of the model

Variable	Coefficient	Std. Error	t	P-value
GS	0.277007	0.127988	2.164317	0.0304
EN	0.422568	0.182652	2.313511	0.0207
_const	-50.59468	20.81568	-2.430604	0.0151

a) Write the logit/ probit model?

b) Estimate the ability for getting job when: GS=70; EN=80.

c) At these grade, if GS increases by 1 unit and EN fixed, how is the change in P(Y=1|X)?

Exercise 2. Estimation of the model

Variable	Coefficient	Std. Error	z-Statistic	Prob
GS	0.155186	0.068907	2.252110	0.0243
EN	0.235860	0.094148	2.505203	0.0122
_const	-28.25491	10.68567	-2.644188	0.0082

a) Write the probit/ logit model?

b) Estimate the ability for getting job when: GS=70; EN=80.

c) At these grade, if GS increases by 1 unit and EN fixed, how is the change in P(Y=1|X)?

Exercise 3: Estimation of logit/probit model. D=1 if male student, D=0 if female student.

Variable	Coefficient	Std. Error	z-Statistic	Prob
GS	0.309411	0.163917	1.887607	0.0591
EN	0.422025	0.235283	1.793693	0.0729
D	2.960152	1.237868	2.391332	0.0168
_const	-53.98806	28.14632	-1.918122	0.0551

a) Write Logit/ Probit

b) Calculate probability of getting job of male and female student with the same (GS, EN) = (70,80)